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MPJ

LAB ASSIGNMENT - 04

* **AIM**: Write a Java program to showcase the use of file I/O Exception Handling in Java.

* **OBJECTIVES**:
  - To study I/O stream classes.
  - Understand the diff. types of exceptions.
  - Exception Handling using various Exception classes.

* **THEORY**:

1) **Java I/O : input-output stream classes**

Java input-output is used to read and write data to files, console or other devices using streams.
A stream is a flow of data.
Types :
1. Byte Stream
2. Character Stream
3. Standard Stream

2) **Exception Handling**

Exception Handling is a mechanism to handle runtime errors so that the program continues execution.

Keywords :
- try → Code that may cause exception
- catch → Handle exception
- finally → Executes always
- throw → Explicitly throw exception

Flow: 1. Code runs inside try.
2. If exception occurs → control goes to catch
3. finally block executes.

3) Types of Exception Classes

i) Checked Exception – Checked at compile time
– Must be handled using try-catch or throws.

ii) Unchecked Exception – Occur at runtime.
– Not compulsory to handle

iii) Errors – Serious problems, not handled by program.

* PLATFORM: Open Source Java Programming tool like Ellipse / NetBeans.

* CONCLUSION: Thus studied the exception handling concepts in Java and implemented usage of I/O system.

* FAQ

(1) Diff. b/w error & exception.

(1)

ERROR	EXCEPTION
→ Conditions that can be handled.	Serious problems.
→ Can be caught using try-catch.	Cannot usually be handled.

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(2) Write a few examples for checked & unchecked exceptions.

(2) Checked Exception:
- IOException
- SQLException
- File Not Found

Unchecked Exception:
- Arithmetic Exception
- Null Pointer Exception
- Array Index Out of Bounds

(3) What are the diff. keywords in Exception Handling?

(3)
- try → Block with risky code
- catch → Handles exception
- finally → Always executes
- throw → Used to explicitly throw
-

(4) Diff. b/w throw & throws.

(4)

Throw	Throws
→ Used inside method.	Used in declaration
→ Throw exception manually	To declare exceptions
→ Single Exception	Multiple.

(5) List down various byte & character stream classes in Java.

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(5) • Byte Stream Classes: i) Input Stream
ii) Output Stream
iii) File Input Stream
iv) File Output Stream
v) Buffered Input Stream.

• Character Stream Classes: i) Reader
ii) Writer
iii) File Reader
iv) File Writer